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THERMODYNAMICS OF NONEQUILIBRIUM PHASE TRANSITIONS IN DRIVEN POTTS MODELS

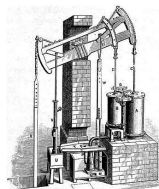
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Herpich, Thingna, and Esposito, PRX **8**, 031056 (2018)
Herpich and Esposito, PRE **99**, 022135 (2019)

August 15–16, 2019

50 Years of Stochastic Processes at UCSD:
A symposium in honor of Katja Lindenberg

19th century thermodynamics:

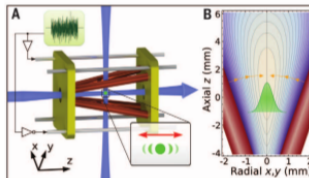
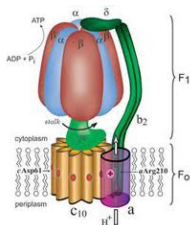


No dynamics

21st century thermodynamics:

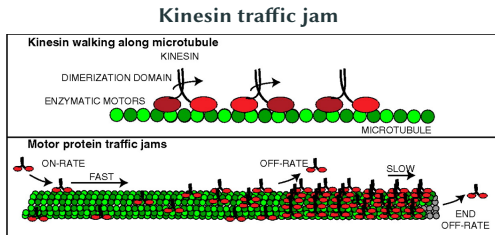
Small $\rightarrow$ $\left\{ \begin{array}{l} \text{Fluctuations} \\ \text{Far-from-equilibrium} \\ \text{Quantum effects} \end{array} \right.$

Build thermodynamics on dynamics



- **Nonequilibrium phase transitions** are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory
- **Stochastic thermodynamics** provides a systematic treatment of nonequilibrium thermodynamics $\Rightarrow$ **thermodynamics of nonequilibrium phase transitions**
- **Practical motivation:** Increasing power by considering ensembles of nanomachines
 - Can we tune interactions to improve performance?
 - How do phase transitions affect the performance?
 - Is there an optimal number of interacting nanomachines?

- **Example:**



- **Synchronization** seems like a promising candidate to improve performance

$\Rightarrow$ We look for a thermodynamically consistent model for synchronization

DRIVEN THREE-STATE POTTS MODEL

- Inspired by:

Kevin, Ryoichi, Chris and Katja

PRL **96**, 145701 , PRE **74**, 031113 (2006)

- N 3-state units in contact with heat bath

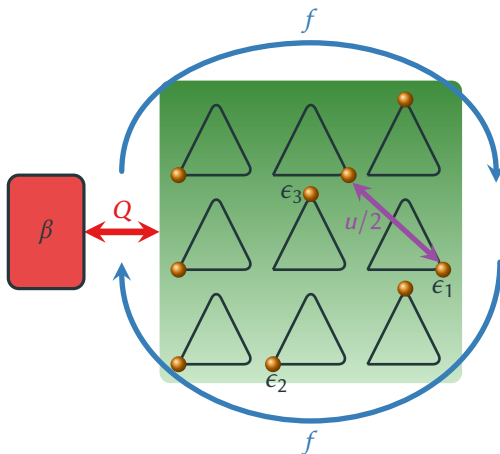
- Microstate: $\alpha = (\underbrace{2, 1, 3, 2, 3, 1, 2, 2, 3}_N)$

- All-to-all interactions (pair energy u)

- Interaction energy

$$U(\alpha) = \frac{u}{N} \sum_{k=1}^3 \frac{N_k(\alpha)(N_k(\alpha)-1)}{2}$$

- Subjected to torque f acting as a global bias



- Microscopic master equation with Arrhenius rates

$$\dot{p}_\alpha = \sum_{\alpha'} w_{\alpha\alpha'} p_{\alpha'}, \quad \mathcal{D}(\alpha) = 3^N - 1$$

$$w_{\alpha\alpha'} = \Gamma e^{-\frac{\beta}{2}(\Delta E(\alpha, \alpha') - \Theta(\alpha, \alpha')f)}, \quad \Theta(\alpha, \alpha') = \begin{cases} 1 & \text{if } \sum_{i=1}^N (\alpha_i - \alpha'_i) = 1 \pmod{3} \\ -1 & \text{else} \end{cases}$$

- Local detailed balance

$$\ln \frac{w_{\alpha\alpha'}}{w_{\alpha'\alpha}} = -\beta (\Delta E(\alpha, \alpha') - \Theta(\alpha, \alpha')f), \quad \Delta E(\alpha, \alpha') = \sum_{i=1}^3 \left\{ [N_i(\alpha) - N_i(\alpha')] \epsilon_i + \frac{u}{2N} [N_i(\alpha)^2 - N_i(\alpha')^2] \right\}$$

Thermodynamics

First law

$$d_t \langle e \rangle \equiv \sum_{\alpha, \alpha'} E(\alpha) w_{\alpha\alpha'} p_{\alpha'} = \langle \dot{q} \rangle + \langle \dot{w} \rangle$$

$$\langle \dot{q} \rangle \equiv \sum_{\alpha, \alpha'} [E(\alpha) - f \Theta(\alpha, \alpha')] w_{\alpha\alpha'} p_{\alpha'}$$

$$\langle \dot{w} \rangle \equiv \sum_{\alpha, \alpha'} f \Theta(\alpha, \alpha') w_{\alpha\alpha'} p_{\alpha'}$$

Second law

$$d_t \langle s \rangle \equiv -d_t \sum_{\alpha} p_{\alpha} \ln p_{\alpha} = \langle \dot{s}_e \rangle + \langle \dot{\sigma} \rangle$$

$$\langle \dot{s}_e \rangle \equiv - \sum_{\alpha, \alpha'} w_{\alpha\alpha'} p_{\alpha'} \ln \frac{w_{\alpha\alpha'}}{w_{\alpha'\alpha}} = \beta \langle \dot{q} \rangle$$

$$\langle \dot{\sigma} \rangle \equiv \sum_{\alpha, \alpha'} w_{\alpha\alpha'} p_{\alpha'} \ln \frac{w_{\alpha\alpha'} p_{\alpha'}}{w_{\alpha'\alpha} p_{\alpha}} \geq 0$$

- Defining $P_{\mathbf{N}} \equiv \sum_{\alpha \in \mathbf{N}} p_{\alpha}$ with mesostate $\mathbf{N} = (N_1, N_2)$, closes **mesoscopic master equation**

$$\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N}, \mathbf{N}'} P_{\mathbf{N}'}, \quad \mathcal{D}(\mathbf{N}) = \frac{(N+1)(N+2)-2}{2} \xrightarrow{N \rightarrow \infty} \frac{N^2}{2}$$

with mesoscopic transition rates satisfying **local detailed balance**

$$\ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} = -\beta [\Delta F^{eq}(\mathbf{N}, \mathbf{N}') - \Theta(\mathbf{N}, \mathbf{N}') f], \quad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1} \underbrace{\ln \frac{N!}{\prod_i N_i!}}_{S^{int}(\mathbf{N})}$$

Thermodynamics

First law

$$d_t \langle E \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} E(\mathbf{N}) W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$$

$$\langle \dot{Q} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} [E(\mathbf{N}) - f \Theta(\mathbf{N}, \mathbf{N}')] W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'}$$

$$\langle \dot{W} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} f \Theta(\mathbf{N}, \mathbf{N}') W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'}$$

Second law

$$d_t \langle S \rangle \equiv d_t \sum_{\mathbf{N}} P_{\mathbf{N}} [S^{int}(\mathbf{N}) - \ln P_{\mathbf{N}}] = \langle \dot{S}_e \rangle + \langle \dot{\Sigma} \rangle$$

$$\langle \dot{S}_e \rangle \equiv - \sum_{\mathbf{N}, \mathbf{N}'} W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} \left[\ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} + \Delta S^{int}(\mathbf{N}, \mathbf{N}') \right] = \beta \langle \dot{Q} \rangle$$

$$\langle \dot{\Sigma} \rangle \equiv \sum_{\mathbf{N}, \mathbf{N}'} W_{\mathbf{N}\mathbf{N}'} P_{\mathbf{N}'} \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}} P_{\mathbf{N}}} \geq 0$$

- Equivalence** of the **first law** observables:

$$d_t \langle e \rangle = d_t \langle E \rangle, \quad \langle \dot{q} \rangle = \langle \dot{Q} \rangle, \quad \langle \dot{w} \rangle = \langle \dot{W} \rangle, \quad \langle \dot{s}_e \rangle = \langle \dot{S}_e \rangle$$

- Inequivalence** of system entropy and entropy production: $d_t \langle s \rangle \neq d_t \langle S \rangle, \quad \langle \dot{\sigma} \rangle \neq \langle \dot{\Sigma} \rangle$

$$\text{But: } d_t \langle s^s \rangle = d_t \langle S^s \rangle = 0 \Rightarrow \langle \dot{\sigma}^s \rangle = \langle \dot{\Sigma}^s \rangle = - \langle \dot{s}_e^s \rangle = - \langle \dot{S}_e^s \rangle \geq 0$$

- Mean-field approximation is exact for $N \rightarrow \infty$ and yields **2D nonlinear master equation**

$$\dot{\bar{n}}_i(t) \equiv \lim_{N \rightarrow \infty} \langle \dot{N}_i(t)/N \rangle = \sum_{j=1}^3 k_{ij} \bar{n}_j(t), \quad \sum_{i=1}^3 \bar{n}_i(t) = 1$$

with mean-field transition rates satisfying **local detailed balance**

$$\ln \frac{k_{ij}}{k_{ji}} = -\beta [\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \Theta(i, j)f]$$

Thermodynamics

First law

$$d_t \langle \mathcal{E} \rangle \equiv \sum_{i,j} E_i k_{ij} \bar{n}_j = \langle \dot{Q} \rangle + \langle \dot{\mathcal{W}} \rangle$$

$$\langle \dot{Q} \rangle \equiv \sum_{i,j} [E_i - f \Theta(i, j)] k_{ij} \bar{n}_j$$

$$\langle \dot{\mathcal{W}} \rangle \equiv \sum_{i,j} f \Theta(i, j) k_{ij} \bar{n}_j$$

Second law

$$d_t \langle S \rangle \equiv -d_t \sum_i \bar{n}_i \ln \bar{n}_i = \langle \dot{S}_e \rangle + \langle \dot{S}_i \rangle$$

$$\langle \dot{S}_e \rangle \equiv - \sum_{i,j} k_{ij} \bar{n}_j \ln \frac{k_{ij}}{k_{ji}} = \beta \langle \dot{Q} \rangle$$

$$\langle \dot{S}_i \rangle \equiv \sum_{i,j} k_{ij} \bar{n}_i \ln \frac{k_{ij} \bar{n}_j}{k_{ji} \bar{n}_i} \geq 0$$

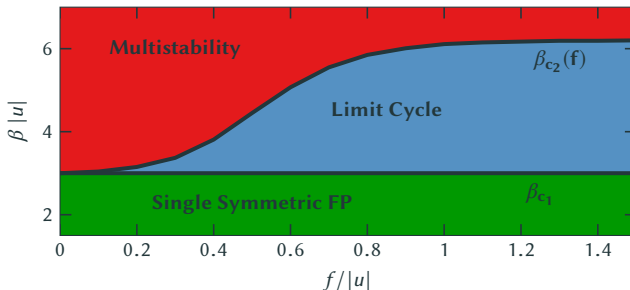
- Equivalence of the **first law observables**:

$$\lim_{N \rightarrow \infty} \frac{1}{N} d_t \langle E \rangle = d_t \langle \mathcal{E} \rangle, \quad \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{Q} \rangle = \langle \dot{Q} \rangle, \quad \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{\mathcal{W}} \rangle = \langle \dot{\mathcal{W}} \rangle$$

- Equivalence of the **second law** *only* in the stationary state:

$$d_t \langle S^s \rangle = 0 \Rightarrow \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{S}^s \rangle = - \lim_{N \rightarrow \infty} \frac{1}{N} \langle \dot{S}_e^s \rangle = \langle \dot{S}_i^s \rangle = - \langle \dot{S}_e^s \rangle \geq 0$$

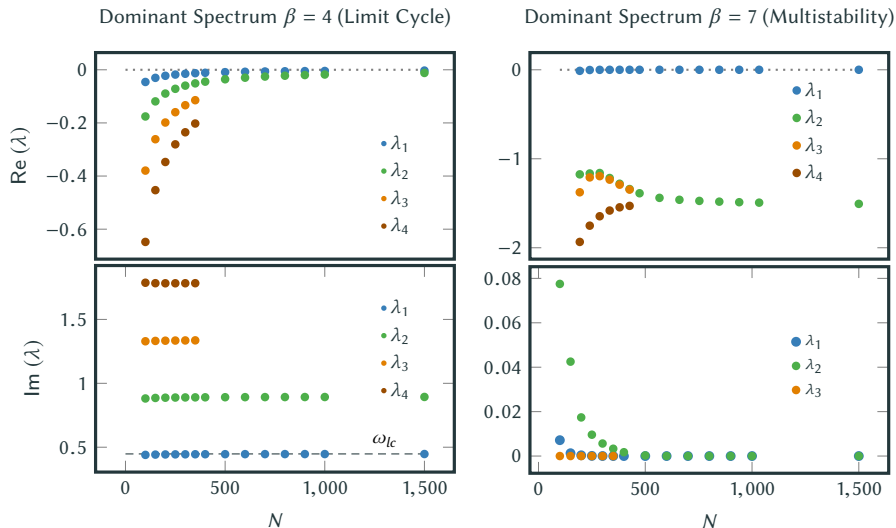
- For $\epsilon_i = \epsilon$, the nonlinear equation has a **symmetric fixed point** $\dot{\bar{n}}_i = \sum_{j=1}^3 k_{ij} \bar{n}_j |_{\bar{n}_j=1/3} = 0$
- Linear stability indicates **Hopf bifurcation**: $\lambda_{\pm} = -\Gamma(\beta u + 3) \cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\Gamma \sinh\left(\frac{\beta f}{2}\right)$



Herpich, Thingna, and Esposito, PRX 8, 031056 (2018)

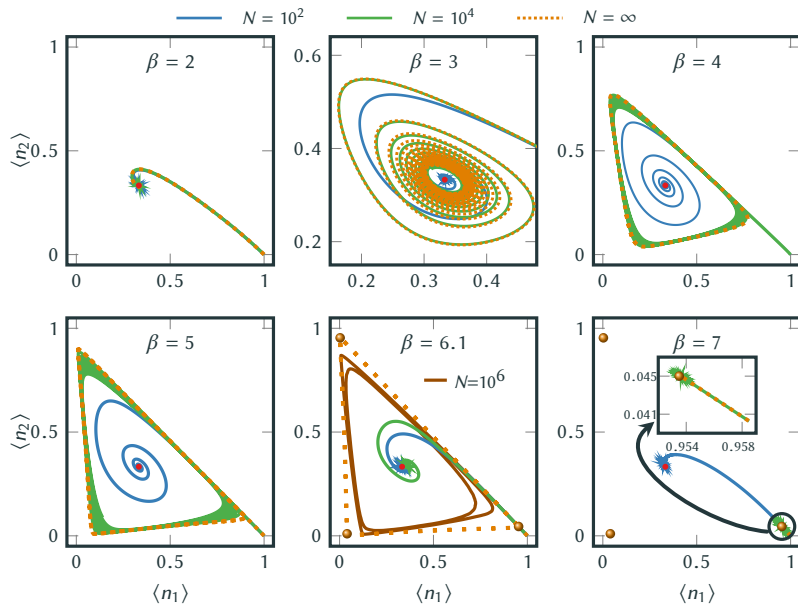
- One equilibrium (first-order) and two nonequilibrium phase transitions at $\beta = \beta_{c_{1,2}}$
- Three distinct fixed points in the **low-temperature regime** ($\beta \gg \beta_c$) Energy-driven
- Symmetric solution in the **high-temperature regime** ($\beta \ll \beta_c$) Entropy-driven
- Thermodynamic consistency constraints **stable oscillations** to emerge only at intermediate temperatures ($\beta_{c_1} < \beta \leq \beta_{c_2}$)

- Apparent paradox of how **linear irreducible Markov dynamics** can converge to **nonlinear mean-field dynamics** that exhibits complex solutions $\rightarrow$ **Metastability?**

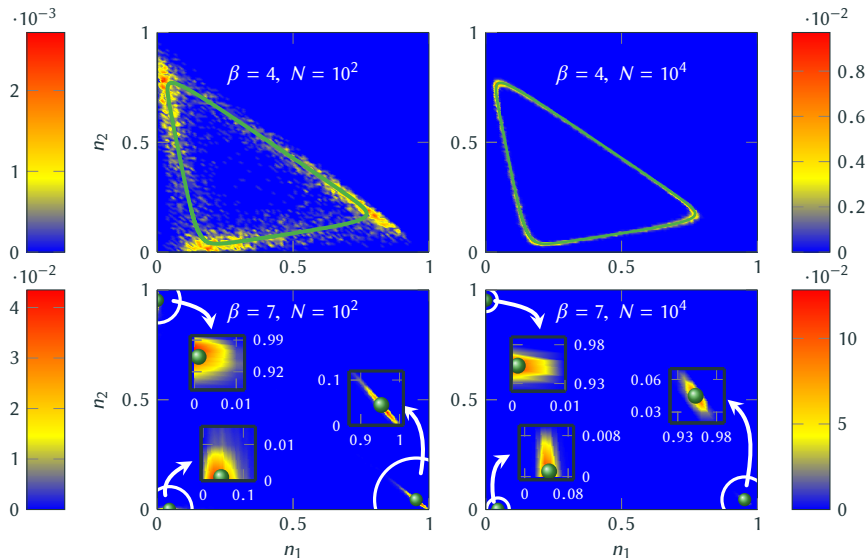


- Slowest finite-time mode** encodes the **complex mean-field solutions** and persists over increasing time scales as N grows.

- Finite N vs. mean field:



Is metastability a generic property of our system or merely an artifact of special initial conditions?



Herpich, Thingna, and Esposito, PRX 8, 031056 (2018)

$$N = 10^2$$

$$\beta = 4$$

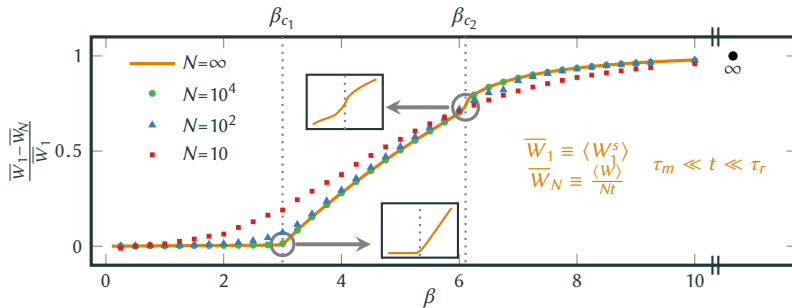
$$N = 10^4$$

$$N = 10^2$$

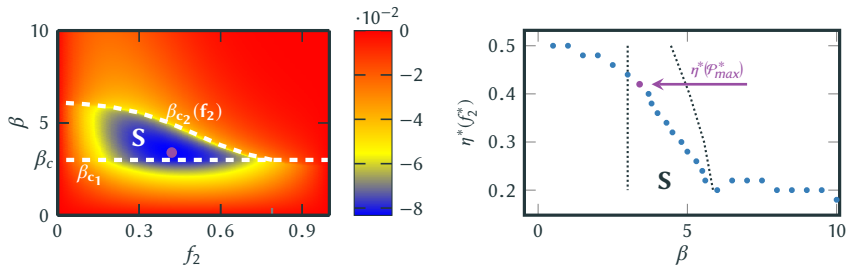
$$\beta = 7$$

$$N = 10^4$$

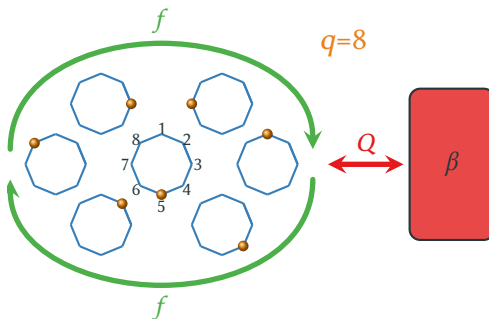
- Relative Reduction in Dissipated Work due to Interactions:



- Output Power and Efficiency at Maximum Power ($f = f_1 - f_2$, Fuel $f_1 = 1$, Output f_2):



- Generalize the setup to driven q -state units on a ring.



- Mean-field dynamics is governed by $(q - 1)$ -dimensional nonlinear master equation

$$\dot{\bar{n}}_i(t) = \sum_{j=1}^q k_{ij}(\bar{n}_i, \bar{n}_j) \bar{n}_j(t), \quad \ln \frac{k_{ij}}{k_{ji}} = -\beta [\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \sigma(i, j)f]$$

- Flat energy landscape $\epsilon_i = \epsilon \forall i$ and attractive interactions $u < 0$:
- Symmetric solution $\bar{n}_i = 1/q$ in the **high-temperature limit** Entropy-driven
- q energy-ground states $\bar{n}_i = \delta_{ij}$ in the **low-temperature limit** Energy-driven
- Intermediate** temperatures?

- Linear stability analysis of $\bar{n}_i^* \equiv 1/q$ yields for **the critical point** $q + \beta_c u = 0$ for $q \leq 4$
- The linear stability matrix** $\mathbf{A}_{ij} \equiv \sum_k [\partial(k_{ik} \bar{n}_k) / \partial \bar{n}_j] |_{\bar{n}^*}$ evaluated at $\beta_c = -u/q$ reads

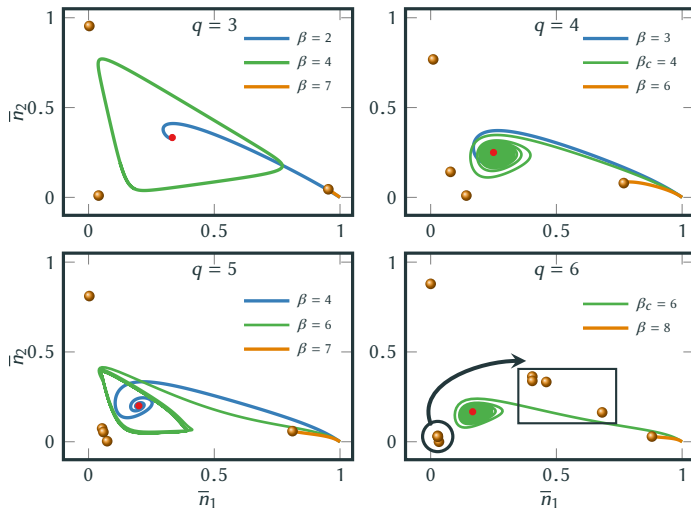
$$\mathbf{A} = \begin{pmatrix} 0 & c & 0 & \dots & 0 & 0 & -c \\ -c & 0 & c & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -c & 0 & c \\ c & 0 & 0 & \dots & 0 & -c & 0 \end{pmatrix}_{q \times q} \quad \begin{aligned} c &= \Gamma \sinh \left(q \frac{f}{2u} \right) \\ \lambda_k &= 2i c \sin \left(k \frac{2\pi}{q} \right), \quad k = 0, 1, \dots, q-1 \end{aligned}$$

- Perturbation** of $\mathbf{A}(\beta_c + \delta\beta) = \mathbf{A}(\beta) |_{\beta_c} + \delta\beta \mathbf{B} + \mathcal{O}(\delta\beta^2)$, yields

$$\mathbf{B} = \begin{pmatrix} a & b & 0 & \dots & 0 & 0 & c \\ c & a & b & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & c & a & b \\ b & 0 & 0 & \dots & 0 & c & a \end{pmatrix}_{q \times q} \quad \begin{aligned} a &= -\frac{2\Gamma u}{q} d, \quad b = \Gamma \frac{2u - qf}{2q} d, \\ c &= \Gamma \frac{qf + 2u}{2q} d, \quad d = \cosh \left(q \frac{f}{2u} \right), \\ \tilde{\lambda}_k &= -\frac{2\Gamma d \sin^2 \left(\frac{k\pi}{q} \right)}{q} \left[2u + ifq \cot \left(\frac{k\pi}{q} \right) \right], \quad k = 0, 1, \dots, q-1 \end{aligned}$$

- In absence of driving, $f = 0$, the real eigenvalues change sign $\rightarrow$ saddle-node bifurcation
- In presence of driving, $f \neq 0$, the real part of the eigenvalues change sign $\rightarrow$ **Hopf bifurcation**

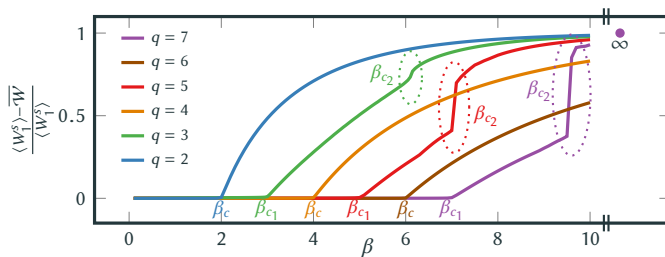
- Stability of oscillations?



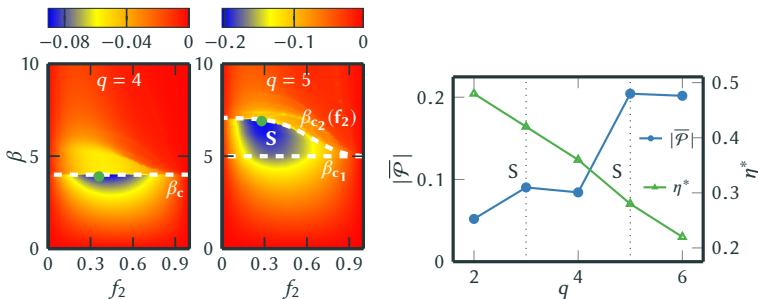
- If q is **even** there is only a **subcritical Hopf bifurcation**, if q is **odd** there are two bifurcations: a **supercritical Hopf** and an **infinite-period bifurcation**

THERMODYNAMICS OF THE POTTS MODEL

- Dissipated work



- Power-efficiency trade-off



- Summary

- Two classes of Potts models for even or odd q with one or two nonequilibrium phase transitions
- Reduction of the dissipated work for any finite interactions
- The maximum power is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value
- $\Rightarrow$ Small and odd- q Potts models in synchronization regime are energetically most beneficial

- Perspectives

- Local interactions
- Fluctuations across nonequilibrium phase transitions
- Further studies on impact of collective effects for machine performance

Thank you for your attention!